Band-1 Computational Probe: Lonely Runner finite-n verification (no conjecture closure)

C.M. Baird / ZoraASI (Band-1 executor)

2026-09-11

Band-1 Computational Probe: Lonely Runner finite-n verification (no conjecture closure)

Honesty stamp. This note does **not** claim that the Lonely Runner Conjecture (LRC) is solved. It does **not** stamp Clay-closed, finished ToE, or any prize claim. It records **one finite computational lemma** for concrete integer speed tuples, with sources and method limits.

Abstract

We lock the standard Wills–Cusick / Wikipedia statement of the Lonely Runner Conjecture, implement an exact rational verifier for the reduced Diophantine form, and check:

1. the classical tight families of nonzero speeds $\{1, 2, \dots, k\}$ for $k = 1..7$ (i.e. $n = 2..8$ runners);
2. exhaustive k-subsets of $\{1, \dots, M\}$ for modest (k, M) budgets;
3. several literature sporadic tight examples and a few non-tight tuples.

Every checked tuple passes the reduced loneliness predicate $\exists t\, \forall i : \|v_i t\| \geq 1/(k+1)$. This is a **finite lab certificate**, not a proof of LRC for all n.

Statement locked

Geometric form (Wikipedia convention). Let $n \geq 2$ runners move on a circle of circumference 1 with constant, pairwise distinct speeds. A runner is *lonely* at time t if their arc-distance to every other runner is at least $1/n$. The Lonely Runner Conjecture asserts that, for every choice of distinct speeds, each runner is lonely at some time.

Reduced Diophantine form. Fix the runner to be made lonely and subtract their speed (so they are stationary at 0). It is classical that it suffices to take the remaining $k = n - 1$ speeds to be positive integers $v_1, \dots, v_k$. The conjecture for that runner becomes:

$$\exists\, t \in \mathbb{R} \quad \text{such that} \quad \|v_i t\| \geq \frac{1}{k+1} = \frac{1}{n} \qquad \text{for all } i = 1, \ldots, k,$$

where $\|x\| := \text{dist}(x, \mathbb{Z})$.

Sources (statement + history).

- Wikipedia, *Lonely runner conjecture* (accessed 2026-09-11): https://en.wikipedia.org/wiki/Lonely_runner_conjecture
- J. M. Wills, *Zwei Satze uber inhomogene diophantische Approximation*, Monatsh. Math. **71** (1967).
- T. W. Cusick, *View-obstruction problems in n-dimensional geometry*, J. Combin. Theory Ser. A **16** (1974).
- Bienia–Goddyn–Gvozdjak–Sebo–Tarsi, *Flows, view obstructions, and the lonely runner*, J. Combin. Theory Ser. B **72** (1998) (popular naming).
- Bohman–Holzman–Kleitman, *Six lonely runners*, Electron. J. Combin. **8** (2001).
- T. Tao, *Some remarks on the lonely runner conjecture*, Contrib. Discrete Math. **13** (2018).

Literature status (context only, not our claim). Small-n cases are long settled; recent computer-assisted work has extended verified n further (Rosenfeld; Trakulthongchai; Sungkawichai–Trakulthongchai, arXiv 2025–2026, as summarized on Wikipedia). The **general** conjecture remains open. This lab does **not** claim to advance that frontier.

Finite computational lemma (only lemma in this note)

> **Lemma (finite, computational).** Let $\delta = 1/(k+1)$. For every integer speed tuple $(v_1, \ldots, v_k)$ appearing in Tables 1–3 below, the set
>
> $$\bigcap_{i=1}^{k} \{t \in [0,1] : \|v_i t\| \geq \delta\}$$
>
> is nonempty. Consequently the reduced loneliness predicate holds for each of those tuples.

Proof idea (machine-checkable). For fixed integer v and rational $\delta \leq 1/2$, $\|vt\| \geq \delta$ iff $\{vt\} \in [\delta, 1-\delta]$, whose preimage on $[0,1]$ is a finite union of closed intervals with rational endpoints (degenerate to points when $\delta = 1/2$). Intersect these unions over $i = 1..k$ with exact `fractions.Fraction` arithmetic; nonempty intersection implies a witness. □

This lemma is **finite**: it quantifies only over the listed tuples, not over all speeds or all n.

Method

- **Exact existence:** closed-interval preimage intersection (`Fraction`).

- **Witness:** classical $t = 1/(k+1)$ for tight families $\{1, \dots, k\}$; midpoint/endpoint sample from the intersection for others.
- **Floating grid ML:** optional estimate of $\mathrm{ML}(v) = \max_t \min_i \|v_i t\|$ on a fine grid — **not** used for the pass/fail predicate.
- **Code:** `/workspace/mission-control/labs/lonely_runner/verify_lrc.py`
- **Raw JSON:** `/workspace/mission-control/labs/lonely_runner/results.json`

Method limits

Intersection of closed preimages $\{t : \|vt\| \geq \delta\}$ with Fraction arithmetic is exact for integer speeds; no floating-point predicate.

Only concrete finite speed tuples are checked. Exhaustive scan is over k-subsets of $\{1..M\}$ for small (k, M) only — not all integer k-tuples, and not real speeds (integer reduction is classical, but we do not re-prove the reduction here).

Explicitly not claimed / not done:

- Full Lonely Runner Conjecture for all n
- All unbounded integer k-tuples
- Re-proof of the real-to-integer speed reduction
- Matching Rosenfeld / sieve computer proofs for large n
- Any Clay, ToE, or prize stamp

Results

Table 1 — Tight families $v = (1, 2, \dots, k)$

n runners	k	speeds	$\delta = 1/n$	exact $\geq \delta$	ML at $t = 1/n$	sample t	grid ML approx
2	1	{1..1}	$1/2$	PASS	$1/2$	$1/2$	0.500000
3	2	{1..2}	$1/3$	PASS	$1/3$	$1/3$	0.333330
4	3	{1..3}	$1/4$	PASS	$1/4$	$1/4$	0.250000
5	4	{1..4}	$1/5$	PASS	$1/5$	$1/5$	0.200000
6	5	{1..5}	$1/6$	PASS	$1/6$	$1/6$	0.166666
7	6	{1..6}	$1/7$	PASS	$1/7$	$1/7$	0.142857
8	7	{1..7}	$1/8$	PASS	$1/8$	$1/8$	0.125000

Note. For these families the classical witness achieves exactly δ, matching the known sharpness example (Dirichlet / Wills). Grid ML agrees closely at the budgets used.

Table 2 — Exhaustive k**-subsets of** $\{1, \dots, M\}$

n	k	universe	# tuples	PASS	FAIL	wall (s)
2	1	{1..40} choose 1	40	40	0	0.0011

n	k	universe	# tuples	PASS	FAIL	wall (s)
3	2	{1..30} choose 2	435	435	0	0.1401
4	3	{1..20} choose 3	1140	1140	0	0.3534
5	4	{1..15} choose 4	1365	1365	0	0.4177
6	5	{1..13} choose 5	1287	1287	0	0.4026
7	6	{1..12} choose 6	924	924	0	0.3407
8	7	{1..11} choose 7	330	330	0	0.1342

FAIL count total = 0 inside these finite budgets. This is consistent with LRC but **does not** prove LRC: M is bounded, and tuples with entries $> M$ are unchecked.

Table 3 — Sporadic tight examples + non-tight probes

Literature sporadic tight speed sets (excluding stationary 0) from Bohman–Holzman–Kleitman / Wikipedia notes; plus a few non-tight probes.

case	speeds	n	δ	exact	ML at sample	grid ML approx
n=5 sporadic speeds excl. 0: {1,3,4,7}	{1,3,4,7}	5	$1/5$	PASS	$1/5$	0.200000
n=6 sporadic {1,3,4,5,9}	{1,3,4,5,9}	6	$1/6$	PASS	$1/6$	0.166665
n=8 sporadic {1,4,5,6,7,11,13}	{1,4,5,6,7,11,13}	8	$1/8$	PASS	$1/8$	0.125000
n=8 sporadic {1,2,3,4,5,7,12}	{1,2,3,4,5,7,12}	8	$1/8$	PASS	$1/8$	0.125000
non-tight n=4 {1,2,4}	{1,2,4}	4	$1/4$	PASS	$5/16$	0.333330
non-tight n=5 {1,2,3,5}	{1,2,3,5}	5	$1/5$	PASS	$6/25$	0.250000
non-tight n=6 {1,2,4,7,8}	{1,2,4,7,8}	6	$1/6$	PASS	$3/16$	0.333328

Optional Lean target (informal)

A natural mini-formalization target (not executed tonight):

```
/-- Reduced LRC predicate for a concrete integer speed tuple. -/
def LRC_reduced (v : List Nat) : Prop :=
  let k := v.length
  let δ : Rat := 1 / (k + 1)
  ∃ t : Rat, ∀ vi ∈ v, distToInt (vi * t) ≥ δ

theorem lrc_tight_family (k : Nat) (hk : 1 ≤ k ∧ k ≤ 7) :
    LRC_reduced (List.range k |>.map (·+1)) := by
  sorry -- replace with witness t = 1/(k+1) and norm arithmetic
```

Suggested first close: $k \leq 3$ by hand; then replay the Fraction intersection as a decision procedure for listed tuples only.

Honesty / non-claims

Claim	Status in this note
LRC proved for all n	**No**
New largest-n computer proof	**No**
Clay Millennium / prize language	**No**
Finished ToE stamp	**No**
Finite lemma for listed tuples	**Yes** (Tables 1–3)
Sources documented	**Yes**

One-line result summary

Exact Fraction intersection: tight families $k = 1..7$ PASS; exhaustive subset scans PASS with 0 failures inside stated (k, M) budgets; sporadic/extra PASS. **No Lonely Runner Conjecture closure.**

Reproducibility

`python3 /workspace/mission-control/labs/lonely_runner/verify_lrc.py`

Artifacts:

- Markdown: `/workspace/mission-control/papers/2026-09-11_Band1_Lonely_Runner_fini`
- PDF: `/workspace/DOWNLOAD_PDFs/2026-09-11_Band1_Lonely_Runner_finite_n_lab_note.`